Q1. Load the given retail sales dataset into a Pandas DataFrame and display the first 5 rows along with the dataset information to understand its structure.

Python Program:

```
import pandas as pd# Load the retail sales datasetdf = pd.read_csv("retail_sales.csv")# Display the first 5 rowsprint("First 5 Rows of the Dataset:")print(df.head())# Display dataset informationprint("\nDataset Information:")df.info()
```

Explanation

1. `import pandas as pd` – Imports the Pandas library for data analysis.
2. `pd.read_csv("retail_sales.csv")` – Loads the CSV file into a Pandas DataFrame.
3. `df.head()` – Displays the first five rows of the dataset.
4. `df.info()` – Displays information about the dataset, including:
    - Number of rows and columns
    - Column names
    - Data types (integer, float, object, etc.)
    - Number of non-null values
    - Memory usage

Expected Output

First 5 Rows: The output displays the first five records from the retail sales dataset, including its available columns.

Dataset Information: The output provides the number of entries, column names, non-null counts, data types, and memory usage.

Note: Replace `"retail_sales.csv"` with the actual file path if your dataset has a different name or location.

Q2. Inspect the dataset and identify missing values in each column. Display the total number of missing values present in the dataset.

Python Program:

```
import pandas as pd# Load the retail sales datasetdf = pd.read_csv("retail_sales.csv")# Identify missing values in each columnprint("Missing Values in Each Column:")print(df.isnull().sum())# Display total number of missing valuestotal_missing = df.isnull().sum().sum()print("\nTotal Missing Values in Dataset:", total_missing)
```

Explanation

1. `df.isnull()` – Checks each cell for missing values (`NaN`).
2. `df.isnull().sum()` – Counts missing values in each column.
3. `df.isnull().sum().sum()` – Calculates the total number of missing values across the entire dataset.

Expected Output

The output will depend on your dataset. For example:

- Missing Values in Each Column:
- OrderID 0
- Product 2
- Quantity 1
- Price 0
- Sales 3
- dtype: int64
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Total Missing Values in Dataset: 6

Note: The output above is illustrative; actual counts may differ.

Conclusion: This analysis helps identify columns containing incomplete data so that appropriate data-cleaning techniques can be applied before further analysis.

Q3. Replace missing values in the Quantity and Price columns with their respective mean values using Pandas.

Python Program:

```
import pandas as pd# Load the retail sales datasetdf = pd.read_csv("retail_sales.csv")# Replace missing values with the mean of each columndf["Quantity"] = df["Quantity"].fillna(df["Quantity"].mean())df["Price"] = df["Price"].fillna(df["Price"].mean())# Display the updated datasetprint("Dataset after Replacing Missing Values:")print(df.head())# Check missing values in Quantity and Priceprint("\nMissing Values After Replacement:")print(df[["Quantity", "Price"]].isnull().sum())
```

Explanation

1. `df["Quantity"].mean()` calculates the average of the non-missing values in the Quantity column.
2. `df["Quantity"].fillna()` replaces missing Quantity values with that average.
3. The same process is applied to the Price column.
4. `isnull().sum()` verifies whether any missing values remain in these two columns.

Expected Output

- Missing Values After Replacement:
- Quantity 0
- Price 0

dtype: int64

Note: This output assumes both columns contain at least one valid numeric value.

Conclusion: Missing values in the Quantity and Price columns are replaced with their respective means, helping retain records for further data analysis.

Q4. Remove all rows where the Product Category or Region column contains missing values.

Python Program:

```
import pandas as pd# Load the retail sales datasetdf = pd.read_csv("retail_sales.csv")# Display the number of rows before removalprint("Rows before removal:", len(df))# Remove rows where Product Category or Region has missing valuesdf = df.dropna(subset=["Product Category", "Region"])# Display the updated datasetprint("\nDataset after removing missing values:")print(df.head())# Display the number of rows after removalprint("\nRows after removal:", len(df))# Check missing values in the specified columnsprint("\nMissing values:")print(df[["Product Category", "Region"]].isnull().sum())
```

Explanation

1. `dropna(subset=["Product Category", "Region"])` removes rows where either the Product Category or Region column contains a missing value.
2. `len(df)` displays the number of rows before and after removal.
3. `isnull().sum()` verifies that no missing values remain in these two columns.

Conclusion: Rows with missing values in Product Category or Region are removed, improving dataset completeness for reliable analysis. Other columns are not used as criteria for row removal.

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